This assignment will not be graded and is only for practice.

1) Choose the set of correct options. ***1 point***

- [ ] Suppose $\beta = \{v_1, v_2, \ldots, v_n\}$ is an orthogonal basis of an inner product space V. If there exists some $v \in V$, such that $\langle v, v_i \rangle = 0$ for all $i = 1, 2, \ldots, n$, then $v = 0$.
- [ ] There exists an orthonormal basis for $\mathbb{R}^n$ with the standard inner product.
- [ ] If P_W denotes the linear transformation which projects the vectors of an inner product space V to a subspace W of V, then $range(P_W) \cap null\ space(P_W) = \{0\}$, where 0 denotes the zero vector of V.
- [ ] $\begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$ cannot represent a matrix corresponding to some projection.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Suppose $\beta = \{v_1, v_2, \ldots, v_n\}$ is an orthogonal basis of an inner product space V. If there exists some $v \in V$, such that $\langle v, v_i \rangle = 0$ for all $i = 1, 2, \ldots, n$, then $v = 0$.
There exists an orthonormal basis for $\mathbb{R}^n$ with the standard inner product.
If P_W denotes the linear transformation which projects the vectors of an inner product space V to a subspace W of V, then $range(P_W) \cap null\ space(P_W) = \{0\}$, where 0 denotes the zero vector of V.
$\begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$ cannot represent a matrix corresponding to some projection.

Solution:

Before we proceed to the solution you may want to recall the following weekly contents:

- Watch : Week 7 Lecture 4, Week 8 Lectures 1,2,3,4
- Solve : Practice assignment: Questions 2 and 3

Option 1:

$\beta = \{v_1, v_2, \ldots, v_n\}$ is an orthonormal basis of the inner product space V.
Step 1: As $v \in V$, we can express v as a linear combination of the vectors $v_1, v_2, \ldots, v_n$ as follows:
$v = \alpha_1 v_1 + \ldots + \alpha_n v_n$, where $\alpha_1, \ldots, \alpha_n \in \mathbb{R}$
Taking inner product of v with v_1, we have $\langle v, v_1 \rangle = \langle \alpha_1 v_1 + \ldots + \alpha_n v_n, v_1 \rangle$
Recall two properties of inner products on V:
i) $\langle u_1 + u_2, u_3 \rangle = \langle u_1, u_3 \rangle + \langle u_2, u_3 \rangle$ for any vectors $u_1, u_2, u_3 \in V$.
ii) $\langle cu_1, u_2 \rangle = c\langle u_1, u_2 \rangle$ for any vectors $u_1, u_2 \in V$ and $c \in \mathbb{R}$.
Using these properties we have,
$\langle v, v_1 \rangle = \alpha_1 \langle v_1, v_1 \rangle + \alpha_2 \langle v_2, v_1 \rangle + \ldots + \alpha_n \langle v_n, v_1 \rangle$.

2) What is the value of $\langle v_1, v_1 \rangle$?

No, the answer is incorrect.
Score: 0

Accepted Answers:

(Type: Numeric) 1

1 point

3) What is the value of $\langle v_2, v_1 \rangle$?

No, the answer is incorrect.
Score: 0

Accepted Answers:

(Type: Numeric) 0

1 point

4) What is the value of $\langle v_i, v_1 \rangle$, for any $i \in \{2, 3 \ldots, n\}$?

No, the answer is incorrect.
Score: 0

Accepted Answers:

(Type: Numeric) 0

1 point

Feedback: Recall the definition of an orthonormal basis and also observe that β is an orthonormal basis.

5) If $\langle v, v_1 \rangle = 0$, then what is the value of α_1?

No, the answer is incorrect.
Score: 0

Accepted Answers:

(Type: Numeric) 0

1 point

Feedback: Observe that $\langle v, v_1 \rangle = \alpha_1$
Similarly we can conclude that, $\alpha_i = 0, \text{ for each } i = 1, 2, \ldots, n$
Hence, $v = 0$.
Option 2: Let $\gamma = \{v_1, v_2, \ldots, v_n\}$ be a given ordered basis of a vector space. The steps in the process are described below to yield an orthonormal basis $\beta = \{u_1, u_2, \ldots, u_n\}$.
Step-1 $w_1 = v_1$, and $u_1 = \frac{w_1}{\|w_1\|}$.
Step-2 $w_2 = v_2 - \langle v_2, u_1 \rangle u_1$ and $u_2 = \frac{w_2}{\|w_2\|}$.
Step-3 $w_3 = v_3 - \langle v_3, u_1 \rangle u_1 - \langle v_3, u_2 \rangle u_2$ and $u_3 = \frac{w_3}{\|w_3\|}$.

$\vdots$

Step-i $w_i = v_i - \sum_{j=1}^{i-1} \langle v_i, u_j \rangle u_j$ and $u_i = \frac{w_i}{\|w_i\|}$.

$\vdots$

Step-n $w_n = v_n - \sum_{j=1}^{n-1} \langle v_n, u_j \rangle u_j$ and $u_n = \frac{w_n}{\|w_n\|}$.

Hence, Option 2 is correct.

Option 3: The projection of a vector to a subspace: Let V be an inner product space, $v \in V$ and $W \subseteq V$ be a subspace. Then the projection of v onto W is the vector in W, denoted by $P_W(v)$, computed as follows:

$$\text{Find an orthonormal basis } \{v_1, v_2, \ldots, v_n\} \text{ for } W.$$
$$\text{Then } P_W(v) = \sum_{i=1}^{n} \langle v, v_i \rangle v_i$$

Step 1:

$$\begin{aligned} P_W^2(v) &= P_W(P_W(v)) \\ &= P_W\left(\sum_{i=1}^{n} \langle v, v_i \rangle v_i\right) \\ &= \sum_{i=1}^{n} P_W(\langle v, v_i \rangle v_i) \\ &= \sum_{i=1}^{n} \langle v, v_i \rangle P_W(v_i) \end{aligned}$$

6) At this point can you identify $P_W(v_i)$? ***1 point***

- ○ v_i
- ○ 0

No, the answer is incorrect.
Score: 0

Accepted Answers:

v_i

Feedback: As v_i is itself in W, the projection of it onto W is nothing but the vector v_i itself. On applying the definition, we get, $P_W(v_i) = \sum_{j=1}^{n} \langle v_i, v_j \rangle v_j = v_i$ as $\langle v_i, v_j \rangle = 0$ when $i = j$ and $\langle v_i, v_i \rangle = 1$.
Hence we have

$$P_W^2(v) = \sum_{i=1}^{n} \langle v, v_i \rangle P_W(v_i) = \sum_{i=1}^{n} \langle v, v_i \rangle v_i = P_W(v)$$

So we can conclude $P_W^2 = P_W$.

Step 2: Let $v \in range(P_W) \cap null\ space(P_W)$. So $v \in range(P_W)$ and $v \in null\ space(P_W)$. As $v \in range(P_W)$ there exists $v' \in V$ such that $P_W(v') = v$, and as $v \in null\ space(P_W)$, we have $P_W(v) = 0$.

$$\begin{aligned} P_W(v') &= v \\ P_W^2(v') &= P_W(P_W(v')) \\ P_W(v') &= P_W(v)[\text{as}, P_W^2 = P_W] \\ P_W(v') &= 0 \\ v &= 0 \end{aligned}$$

Hence $range(P_W) \cap null\ space(P_W) = \{0\}$.

Option 4: If a matrix A represents a projection, then A^2 must be A.

Let $A = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$.

7) Is $A^2 = A$? **1 point**

- ◯ Yes
- ◯ No

No, the answer is incorrect.

Score: 0

Accepted Answers:

No

Hence, $\begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$ cannot represent a matrix corresponding to some projection.

Your score is: 0/7